

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

reasoning process here

power dissipation formula is $P = IV$, and power formula is $P = V^2 / R$.

since the bulb is directly connected to the battery , the voltage is 12 V , and power is 40 W , we can find the current using $P = IV$, $I = P / V$.

then , we can find the resistance using $V = IR$, $R = V / I$, and V is 12 V .

</think>

<answer>

C

</answer>

